



NATIONAL OPEN UNIVERSITY OF NIGERIA

PLOT 91, CADASTRAL ZONE, NNAMDI AZIKIWE EXPRESSWAY, JABI-ABUJA

FACULTY OF SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

2024 1 EXAMINATION

COURSE CODE: CIT308

COURSE TITLE: FORMAL METHODS AND SOFTWARE DEVELOPMENT

COURSE CREDIT: 3 UNITS

TIME ALLOWED: 2 HOURS

INSTRUCTION: ANSWER QUESTION 1 AND ANY OTHER THREE QUESTIONS

Q1.

(a) Discuss 4 qualities of an acceptable user interface? **4 marks**

(b) Draw a data flow diagram for online banking system **5 marks**

(c) Construct a truth table for the formula $(\neg(p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow \neg r$ **12 marks**

(d) List 8 reasons why formal methods are useful? **4 marks**

Q2

(a) What are formal methods? **2 marks**

(b) In formal logic, under what circumstances is the following true? $(x < 3 \wedge x > 5) \Rightarrow x = x + 1$? **6 marks**

(c) Differentiate with examples, the two types of test flow? **7 marks**

Q3.

(a) Discuss the system, integration and user acceptance testing? **6 marks**

(b) Mention 3 types of software development models.

3 marks

- (c) List any 6 terminologies used to describe sets relationship?

3 marks

- (d) Outline 6 stages of the SDLC?

3 marks

Q4

- (a) Using a table, discuss the relationship between the integrity level (low, medium and high) and the formal method stages?

12 marks

- (b) Mention 3 criteria that must be met before any 'formal system' can be considered a logic?

3 marks

Q5

- (a) Given three sets $X=\{3, 4, 7, 9\}$; $Y=\{4, 2, 3\}$ and $Z=\{10, 8, 6\}$,
(i) describe and provide the elements of: $(X \cap Y \cap Z) \cup (X \cup Z)$.
(ii) Shade off the set: $(X \cap Y \cap Z) \cup (X \cup Z)$.

12 marks

- (b) A good software must be operational, transitional and maintainable. Clearly distinguish between these terms?

3 marks

Q6.

- (a) Itemise 6 reasons, why software quality is advantageous to an organization.

3 marks

- (b) Enumerate and describe 4 of the software development phases

4 marks

- (c) Why is it important to have precision in the specification of software?

3 marks

- (d) Enumerate possible reasons why formal methods have not become mainstream software development techniques?